

Design and Simulation of FeSi₂-Based Novel Heterojunction Solar Cells for Harnessing Visible and Near-Infrared Light

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Herein, three novel third-generation (3G) solar cells: n-Si/p-FeSi₂/p⁺-Si, n-Si/p-FeSi₂/p⁺-BaSi₂, and n-CdS/p-FeSi₂/p⁺-BaSi₂ based on the orthorhombic iron disilicide (β-FeSi₂) absorber are demonstrated theoretically for multikilowatt photovoltaic (PV) systems and space applications. These cells overcome the complication of producing low voltages (≤ 450 mV) of FeSi₂-based solar cells due to the narrow bandgap (≈ 0.87 eV) of the absorber. Using crystalline silicon (c-Si), cadmium sulfide (CdS), and orthorhombic barium disilicide (β-BaSi₂) as junction partners, effects of parameters such as the thickness, doping and defect densities, band offsets, and temperature are studied systematically by a solar cell capacitance simulator (SCAPS-1D). The highest open-circuit voltage of 958 mV is attained materially with a 300 nm thin absorber. This article renders the optimization of the PV parameters to improve the device performance with power conversion efficiencies (PCEs) of 28.18%, 31.61%, and 38.93% by the three novel npp⁺ approaches compared to the PCEs of 15.78% and 24.96% for the solar cells n-Si/p-FeSi₂ and p-Si/i-FeSi₂/n-Si, respectively.

1. Introduction

The semiconducting orthorhombic iron disilicide (β-FeSi₂) has been recognized as an eco-friendly green semiconductor material for preparing third-generation (3G) solar cells after silicon (Si) and gallium arsenide (GaAs)-based solar cells.^[1] The second-generation (2G) representative cadmium telluride (CdTe) and copper indium gallium diselenide (CIGS) thin film solar cells (TFSCs) have a good absorption of visible light and give a high power conversion efficiency (PCE) with cost effectiveness.^[2] However, cadmium (Cd) is a heavy metal with toxicity, whereas

gallium (Ga) and indium (In) are scarcely available in the earth's crust.^[3] Considering these environmental and scarcity issues, the scientific community is still searching for new types of solar cells at this present 3G era of solar cell technology.

FeSi₂ has a direct bandgap energy (E_G) of 0.87 eV,^[4,5] with a high absorption coefficient (α) exceeding $1 \times 10^5 \text{ cm}^{-1}$ at a photon energy of 1 eV,^[1,5–7] which is at worst 50 times greater than that of crystalline-silicon (c-Si).^[8] The forming member elements iron (Fe) and Si of FeSi₂ are among the most abundant components found in the earth's crust.^[7] Thus, FeSi₂ is a very low-cost material that can be used as an active absorber material for making solar photovoltaic (PV) cells. Also, FeSi₂ is nontoxic, chemically stable at a high temperature of 937 °C, and highly resistive to oxidation and humidity.^[1] It

has a long carrier diffusion length of 38 μm .^[4] This abundant material has a specific gravity of 4.93 which is placed between Si (2.33) and GaAs (5.33).^[1] Moreover, FeSi₂ offers high resistance against cosmic and radioactive exposure.^[1] At room temperature (RT), FeSi₂ films can be grown on any solid surfaces,^[8] such as Si or quartz substrates. It is worth noting here that FeSi₂ can absorb photons from the visible as well as the near-infrared (NIR) light spectrum up to a wavelength of 1675 nm.^[6] All these attributes make it suitable for application in 3G solar PV technology to be used in space.^[1] Devices such as solar cells, photosensors, and thermoelectric generators can be manufactured from FeSi₂ and these can be integrated fundamentally on Si large-scale integration (LSI) circuits.^[1] Although FeSi₂ films show n-type conductivity by default,^[7] the conductivity type can be altered by varying the ratio of Si to Fe content. For instance, it becomes p-type when the Si/Fe atomic ratio is 1.5–1.8, whereas it turns into n-type when the ratio is 1.9–2.^[9]

A number of works on FeSi₂-based solar cells have already been published. These include the theoretical^[4–6,10] and experimental approaches.^[7,11–14] A summary of the recent technologies of FeSi₂ film preparation has also been reported.^[1] The reported highest efficiency gained so far is 3.7%,^[7] which is much lower than the theoretical efficiencies of 7–28%^[4,10] for FeSi₂-based solar cell. Different types of defects in the FeSi₂ layer introduced during fabrication are the primary cause of this limitation.

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 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/pssa.201900921>.

DOI: 10.1002/pssa.201900921

The lower performance of FeSi_2 solar cells can be attributed to the very low value of the photogenerated open-circuit voltage (V_{OC}) due to its narrow bandgap of 0.87 eV. Experimentally fabricated cells offered lower V_{OC} in the range 176^[12]–450 mV,^[7] which, as a consequence, dictated the lower PCE. To date, the reported highest theoretical efficiency of this cell is 24.7%, and 27.9% in the pin configuration,^[4] with a c-Si, and an amorphous-silicon (a-Si), partner. Recently, a triple-junction (a-Si:H/ $\mu\text{c-Si}$ / $\beta\text{-FeSi}_2$) solar cell has also been reported.^[15] This review of the literature reveals that FeSi_2 -based pn, np, or pin heterojunction solar cells have had either c-Si or a-Si as the junction partner to date.

In addition, the effect of adding various back-surface field (BSF) layers such as c-Si,^[16,17] vanadium pentoxide (V_2O_5),^[18] antimony telluride (Sb_2Te_3),^[19] copper zinc tin sulfide (CZTS),^[3] copper zinc tin selenide (CZTSe),^[20] and tin sulfide (SnS)^[21,22] on the performance of different heterojunction solar cells has already been reported. More recently, it has been reported that the orthorhombic barium disilicide ($\beta\text{-BaSi}_2$) can be applied as the best BSF layer for performance enhancement in CIGS-based solar cells.^[23] The potential of BaSi_2 for solar cell applications has also been reported by many groups.^[23–29]

In this work, we introduce and numerically assess three novel npp^+ heterojunction solar cells based on an FeSi_2 absorber that incorporate heavily doped c-Si, or BaSi_2 , as the BSF layer. We also use cadmium sulphide (CdS) instead of Si as a window material for the first time in this type of solar cells. We comparatively study the PV parameters such as the short-circuit current density (J_{SC}), open-circuit voltage (V_{OC}), fill factor (FF), PCE (η), and quantum efficiency (QE) for the modeled solar cells. This study demonstrates how the newly designed npp^+

heterojunctions with optimized parameters can overcome the problem of producing low voltages and enhance the PCE as compared to the reported values.

2. Device Structure and Simulation

We started our design and simulation from two previously reported heterojunction solar cells^[2–4] shown in Figure 1a,b. First, we evaluated our simulated solar cell structures 1 and 2 with the reported theoretical and experimental works and then started modeling, simulating, and analyzing the new ones. It should be noted that these two structures were analyzed only to compare them with the newly modeled heterojunction solar cells of type 3–5 of Figure 1c–e, respectively. Cell 1 was modified employing p^+ -type Si as a BSF layer adjacent to the FeSi_2 absorber layer at the rear side of the cell, as shown in Figure 1c. Then, cell 3 was modified by replacing the Si BSF with a p^+ -type BaSi_2 BSF layer as identified as the cell of type 4. Finally, the Si window layer was altered by the well-known CdS window layer, which resulted in the cell of type 5 and which significantly enhanced the PV performance of the devices.

To model the first, third, and fourth cells, a 50 nm thin n-type Si was used as the window layer. In cells 3–5 of the npp^+ structures, a heavily doped 50 nm thin BSF layer of either Si (acceptor doping: $5 \times 10^{19} \text{ cm}^{-3}$) or BaSi_2 (acceptor doping: 10^{19} cm^{-3}) was used. The present investigation confirms that FeSi_2 was used as the only active absorber layer for the designed solar cells.

All the simulations and analysis were conducted by a 1D solar cell capacitance simulator (SCAPS-1D) (version: 3.3.07).^[30]

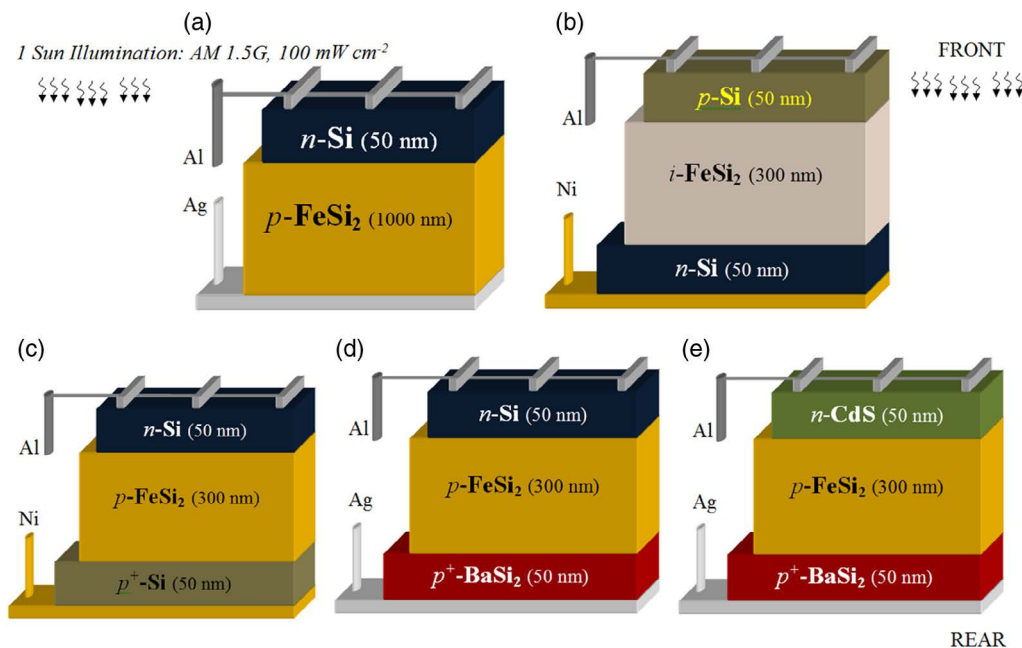


Figure 1. Modeled and optimized heterojunction solar cells: a) cell 1: Al(110)/n-Si/p- FeSi_2 /Ag(100); b) cell 2: Al(110)/p-Si/i- FeSi_2 /n-Si/Ni(110); c) cell 3: Al(110)/n-Si/p- FeSi_2 / $\text{p}^+\text{-Si}$ /Ni(110); d) cell 4: Al(110)/n-Si/p- FeSi_2 / $\text{p}^+\text{-BaSi}_2$ /Ag(100); and e) cell 5: Al(100)/n-CdS/p- FeSi_2 / $\text{p}^+\text{-BaSi}_2$ /Ag(100).

The SCAPS simulator enables the PV research community to analyze 1D devices consisting of a maximum of seven layers. It is a highly declared simulation tool planned to reproduce the electrical characteristics and spectral response of the simulated TFSCs.^[3] In general, it allows studies on tunneling, defects, energetic distribution, generation–recombination (*G–R*) mechanism, etc. This simulation computes energy bands, currents, and concentrations at a given operating condition, *J–V* characteristics, AC characteristics (*C–V*, *C–f*), and spectral responses at light or voltage bias with the help of the Poisson and free-carrier (electron and hole) continuity equations.^[30] It enables PV researchers to numerically evaluate their experimental results in a soft environment.^[20]

All the designed devices were simulated under a 1 sun illumination of 100 mW cm^{-2} and global air mass AM 1.5G solar spectrum at a temperature of 300 K, assuming the ideal values for the series (R_s) and shunt (R_{sh}) resistances. The values for the surface work function (Φ_M) of the contact metals used for collecting the charge carriers were taken from the reported work.^[31] In addition, the charge states and recombinations were considered only for the bulk materials. No defect was assumed at the interfaces between layers. Single acceptor and/or single donor type defects were put in the bulk layer using the Gaussian energetic distribution. **Table 1** shows all the essential parameters of the materials used during the simulation with SCAPS. These electrical, optical, and defect parameters were taken from reliable reports as far as accessible.

3. Results and Discussion

3.1. The Band Diagrams of the Modeled Heterojunctions

Figure 2 a–e show the simulated energy band diagrams for the modeled solar cells. The energy levels in the bands bend at positions near the interface of the differently doped bulk layers. The band diagram is mainly characterized by the energy band gap (E_G), electron affinity (χ), and doping and defect level of the partner layers.

Using the Fermi-level difference, the heavily doped p- and n-type Si layers in the pin structure of cell 2 induce a high drift field within an absorber layer such as the nanocrystalline pin structure.^[6] The value of χ for the FeSi₂ material is assumed to be 3.86 eV^[6] for cell 1 only, whereas it is 4.16 eV^[4] for all other cell types. However, better performance is achieved with the same χ of 4.16 eV in cell 1. For instance, with $\chi = 3.86 \text{ eV}$, the simulation yields the PV parameters $V_{OC} = 389 \text{ mV}$, $J_{SC} = 51.57 \text{ mA cm}^{-2}$, $FF = 76.78\%$, and $\eta = 15.42\%$, whereas with that of 4.16 eV, these parameters are 397 mV, 51.89 mA cm^{-2} , 76.53%, and 15.78%, respectively. Therefore, comparing these values, it can be concluded that the output parameters are almost identical for cell 1 with the aforementioned electron affinities. The change in electron affinities causes spike-like band alignment at the interface, which simply alters the output by a few degrees.

The right-hand side of **Figure 2** shows the band diagrams of the proposed npp⁺ heterojunction solar cells 3–5. In cell 3,

Table 1. Simulation-related materials' data at 300 K from literature.

Parameters	c-Si	CdS	FeSi ₂	BaSi ₂
Layer type	Window/partner/BSF	Window	Absorber	BSF
Conductivity type	n/p/p ⁺	n	p/intrinsic	p ⁺
Thickness [nm] ^{a)}	50	50	300	50
Bandgap, E_G [eV]	1.12 ^[4–6]	2.4 ^[30]	0.87 ^[4,5]	1.3 ^[25,26]
Electron affinity, χ [eV]	4.05 ^[4–6]	4.2 ^[30]	4.16 ^[4]	3.3 ^[26]
Dielectric permittivity, ϵ [relative]	11.9 ^[4–6]	10 ^[30]	22.5 ^[4]	15 ^[25]
Effective CB density, N_C [cm ^{−3}]	2.8×10^{19} ^[4,5]	2.2×10^{18} ^[2,30]	5.6×10^{19} ^[4]	2.6×10^{19} ^[25,27]
Effective VB density, N_V [cm ^{−3}]	1.04×10^{19} ^[4,5]	1.8×10^{19} ^[2,30]	2.08×10^{19} ^[4]	2×10^{19} ^[25,27]
Electron mobility, μ_n [cm ² V ^{−1} s ^{−1}]	1350 ^[4,25]	100 ^[2,30]	100 ^[4]	20 ^[25]
Hole mobility, μ_p [cm ² V ^{−1} s ^{−1}]	500 ^[4,25]	25 ^[2,30]	20 ^[4]	20 ^[25]
Donor concentration, N_D [cm ^{−3}] ^{a)}	$10^{17}/0/0$	10^{18}	0	0
Acceptor concentration, N_A [cm ^{−3}] ^{a)}	$0/10^{17}/5 \times 10^{19}$	0	$10^{17}/0$	10^{19}
Defect type	Acceptor/donor/donor	Acceptor	Donor/both	Donor
Energetic distribution	Gaussian	Gaussian	Gaussian	Gaussian
Peak defect density, $N(t)$ [eV ^{−1} cm ^{−3}] ^{a)}	10^{14}	10^{14}	10^{14}	10^{14}
Characteristic energy [eV]	0.01 ^[4]	0.1 ^[30]	0.03 ^[4,6]	0.01 ^[25]
Reference energy [eV]	0.56	1.2	0.435	0.65
Electron capture cross section for acceptor defect [cm ²]	10^{-17} ^[4,6,25]	10^{-17}	10^{-17} ^[4,6]	—
Hole capture cross section for acceptor defect [cm ²]	10^{-15} ^[4,6,25]	10^{-15}	10^{-15} ^[4,6]	—
Electron capture cross section for donor defect [cm ²]	10^{-15} ^[4,6,25]	—	10^{-15} ^[4,6]	10^{-15} ^[25]
Hole capture cross section for donor defect [cm ²]	10^{-17} ^[4,6,25]	—	10^{-17} ^[4,6]	10^{-17} ^[25]

^{a)}Is a variable field.

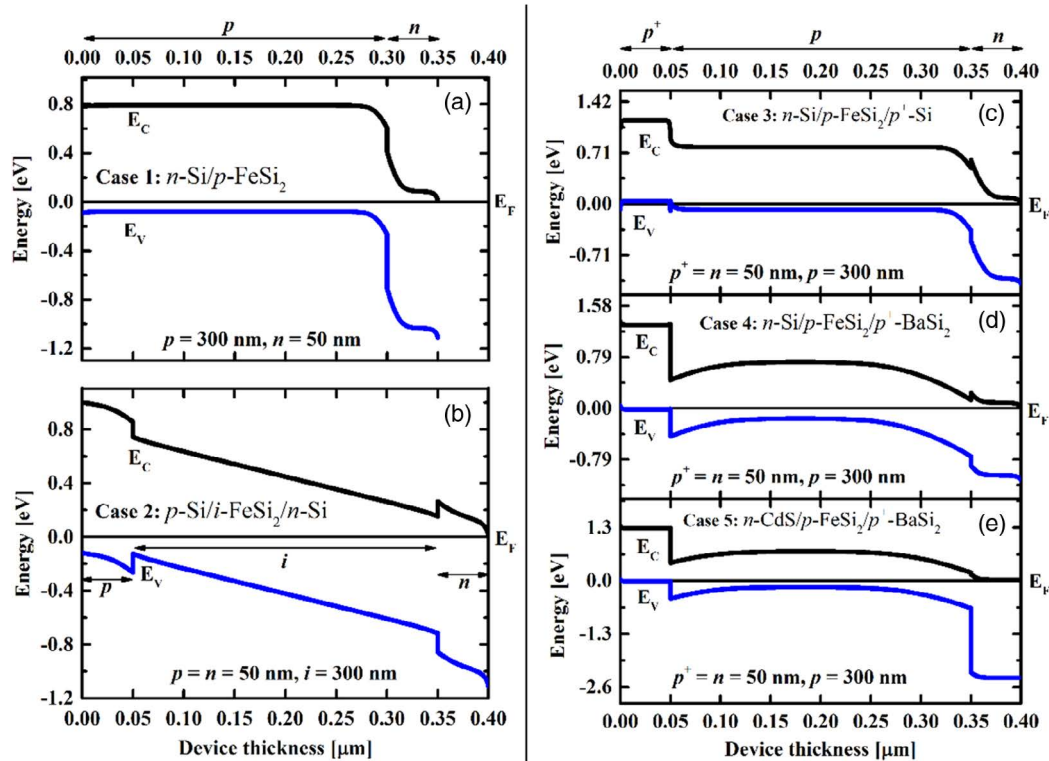


Figure 2. The energy band diagram for the modeled devices at a bias voltage of 0 V.

the Fermi level (E_F) crosses the top of the valence band (VB) energy level (E_V) for the p^+ -type BSF layer due to the heavy doping ($5 \times 10^{19} \text{ cm}^{-3}$) of the layer, resulting in a degenerate semiconductor. In other words, E_F goes inside the E_V level. Again, the difference in conduction energy levels (ΔE_C), i.e., the conduction band (CB) offset (CBO) between the p-type absorber and the n-type window layers, is lower in cells 3–5 compared to that in cell 2. This is why the passage of charge carriers becomes somewhat easier for our proposed cells than cell 2 of the pin design. This, along with the electric field produced by the BSF layer, makes it easier to obtain higher performance by the proposed solar cells compared to the previously reported cells 1 and 2. In the case of cells 1–3, the VB offsets (VBOs) between the p-type absorber and the p^+ -type BSF layer are much smaller than the CBOs between those two layers. For this reason, the BSF and then the back metal easily extract the positive charges (holes) from the absorber, whereas electron extraction is very tough or electrons cannot be extracted at all.^[21] Similarly, entry of holes at the window is prevented by the existing VBO between the absorber and the window layers, but the electron extraction is continued, which eventually improves the PCE of the device.

3.2. The Effect of FeSi₂ Absorber Layer Thickness on the PV Parameters

In this section, the effect of the absorber layer thickness on the solar cell parameters is studied to achieve an optimized thickness. In the meantime, we comparatively investigate the

effect of introducing the BSF layer in cells 3–5 with the BSF-free device of cell 1. The impact of adding various BSF layers on different solar cells has already been reported by many groups.^[3,16–22]

At 300 K, we varied the absorber thickness from 50 nm (0.05 μm) to 2 μm for all cases. For cells 1 and 2, the doping density was 10^{17} cm^{-3} in the 100 nm thin Si layers. Acceptor density, N_A was 10^{17} cm^{-3} in the absorber of cell 1. After the proper investigation of cells 1 and 2, a 50 nm thin window layer was chosen for cells 3 and 4 for the simulation. Cell 3 had the doping densities N_A (FeSi₂) = N_D (Si) = 10^{18} cm^{-3} and N_A (Si) = $5 \times 10^{19} \text{ cm}^{-3}$. The thickness of the p^+ -type BSF layer was 50 and 100 nm in cells 3 and 4, respectively. However, the thickness was 100 nm for the CdS window and 50 nm for the BaSi₂ layer in cell 5. In all cases, the same peak defect density of $10^{14} \text{ eV}^{-1} \text{ cm}^{-3}$ was maintained for all the layers. All these aforementioned values have been taken for varying absorber layer thicknesses in the simulation. **Figure 3** and **4** show the variation of the PV parameters with the change in the absorber layer thickness.

3.2.1. Effect on the Short-Circuit Current (J_{SC})

Because of the high absorption coefficients, a 230 nm thin layer of FeSi₂ was found to be large enough for absorbing all the photons with photon energies greater than the bandgap.^[7] It is reasonable because a very high value of the absorption coefficient makes the absorption depth very low due to the inversely proportional relation. In this work, 100 nm thin FeSi₂ is found

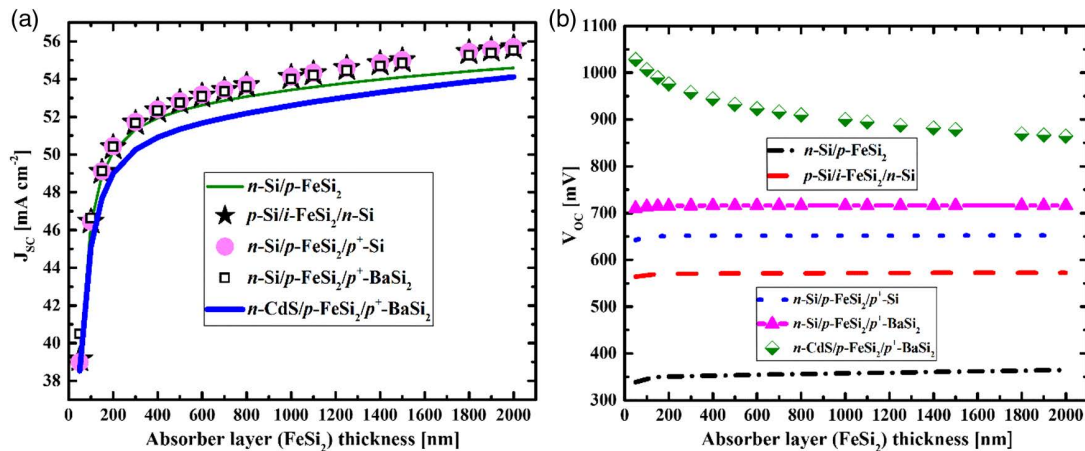


Figure 3. The basic output of the device versus absorber thickness: effect on a) J_{sc} and b) V_{oc} .

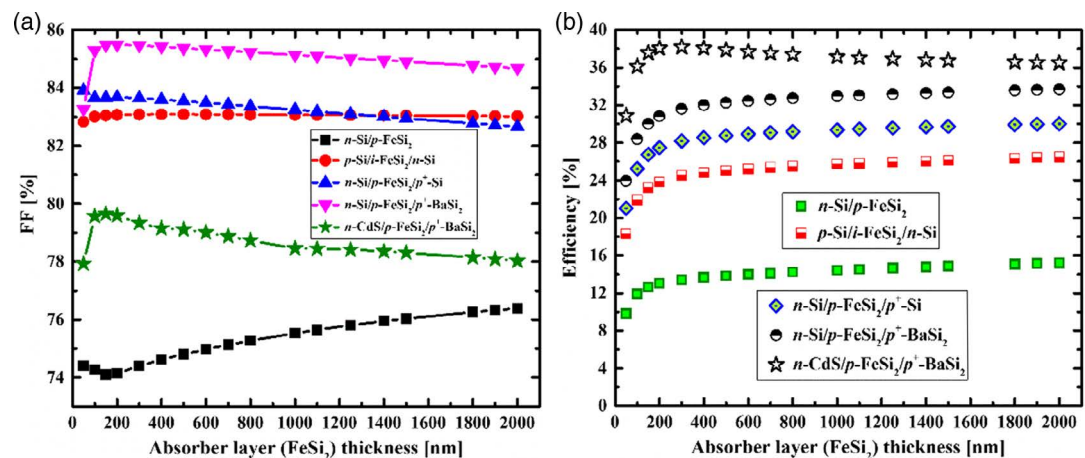


Figure 4. The basic output of the device versus absorber thickness: effect on a) FF and b) PCE.

capable of providing a high J_{sc} from all the five heterojunctions. The photogenerated current increases negligibly after a thickness of 300 nm. With a 300 nm thin absorber, cell 3 offers the highest J_{sc} of 51.73 mA cm^{-2} . Such a high current of 53 mA cm^{-2} was also achieved by the FeSi_2 solar cell.^[6] For FeSi_2 thickness ≥ 300 nm, all the cells provide a minimum J_{sc} of 51 mA cm^{-2} . The lowest possible current of 38.53 mA cm^{-2} is achieved for a 50 nm thin absorber. With increased thickness of the absorber layer, the photogenerated current starts increasing. However, beyond 300 nm, a less than 4 mA increment in J_{sc} occurs by the additional 1700 nm layer. This is also reasonable because the thicker the absorber, the higher is the absorption probability. Another reason for the poorer current at lesser thickness is that the carriers recombine before they can reach the absorber/window junction. A thicker absorber will simply capture more photons and utilize their energy to make additional electron-hole pairs (EHPs). But a thicker absorber needs more materials, and thus costs accordingly. Therefore, manufacturers want optimization of the absorber thickness in a solar cell. With this view, simulation facilitates a lot. Sharing of the applied voltage between the two junctions (n/p and p/p⁺) decreases the dark

current, whereas it increases the light-induced current.^[32] This is shown in Figure 3a using cells 1 and 3.

3.2.2. Effect on the Open-Circuit Voltage (V_{oc})

Absorbing the energies from more photons, the cell produces a higher V_{oc} . At 300 nm thickness, cell 5 offers the highest V_{oc} of 958 mV. The V_{oc} of cell 5 is higher than those of cells 4, 3, 2, and 1 by 243, 307, 373, and 561 mV, respectively. With increased thickness of the absorber, V_{oc} increases for cells 1–4. In contrast, it decreases in case of cell 5. This can be attributed to the associated increase in the dark current with a thicker absorber, which reduces V_{oc} in that cell. It has been reported that increasing the absorber thickness in a BaSi_2/CdS /fluorine-doped tin oxide solar cell causes V_{oc} to decrease gradually.^[24] A very similar decreasing fashion is also observed for a-Si:H solar cells.^[2,33] However, the experimentally fabricated FeSi_2 solar cells gave V_{oc} in the range 176^[12]–450 mV,^[7,13,14] whereas it was 391^[10]–900 mV^[4] for the theoretical works. In this work, the maximum V_{oc} is obtained for cell 5 because the CdS window has a larger bandgap of 2.4 eV^[30] compared to the c-Si

bandgap of 1.12 eV.^[4–6] Similarly, the a-Si window having an E_G of 1.7 eV produced a high V_{OC} of 900 mV, compared to 680 mV by the c-Si window having an E_G of 1.12 eV.^[4]

CdS is the most widely used window material in 2G solar cell technologies,^[2] with the highest transparency (no absorption after 600 nm).^[34] With significant reduction in parasitic absorptions by CdS, i.e., assisting in more photon absorption by the absorber in cell 5, CdS contributes to enhancing V_{OC} more with respect to Si in cell 4. The dark saturation current density (J_0) and the light-induced current density (J_{SC}) are mathematically related to V_{OC} by Equation (1)^[29]

$$V_{OC} \approx \frac{K_B T}{q} \ln \frac{J_{SC}}{J_0} \quad (1)$$

where K_B is Boltzmann's constant ($1.3807 \times 10^{-23} \text{ J K}^{-1}$), q is the elemental charge of an electron ($1.602 \times 10^{-19} \text{ C}$), and T is the absolute temperature (300 K). Thus, V_{OC} has a direct dependence on the ratio of J_{SC} to J_0 . Simulation results reveal that J_0 is much lower in cell 5 than in cell 4. The lower dark current is attributed to the lower parasitic absorption by the CdS window. As J_{SC} is almost the same for cells 4 and 5, the reduced J_0 increases V_{OC} in cell 5 markedly.

The built-in potential, $V_{p/n}$ at the p/n junction of cell 1 can be written as Equation (2)^[32,35]

$$V_{p/n} = \frac{KT}{q} \ln \frac{N_A N_D}{n_i^2} \quad (2)$$

With a BSF layer (heavily doped with hole) having doping density N_a and the p⁺/p junction of the devices in cells 3–5, the built-in potential, $V_{p+/p}$, can be written as Equation (3)^[32,35]

$$V_{p+/p} = \frac{KT}{q} \ln \frac{N_a}{N_A} \quad (3)$$

Summing the built-in potentials for those two junctions, the resultant built-in potential (V_{bi}) increases the V_{OC} of the resulting device with the BSF layer. The band diagram in Figure 1e confirms the presence of degenerate BaSi₂ and CdS layers in cell 5 due to the heavy doping used in the simulation for these layers. This can be attributed to the existing E_F level within the E_V and E_C levels in the BaSi₂ and CdS regions, respectively. Based on the method described in the study,^[36] $V_{p/n}$ and $V_{p+/p}$ are determined for cell 5 using the C–V panel of the SCAPS simulator. This calculation gives the V_{bi} value of $\approx 1.06 \text{ eV}$ for cell 5, which is higher than the E_G of 0.87 eV of the FeSi₂ absorber layer. The value of the built-in potential, V_{bi} , can be greater than the bandgap E_G of the absorber layer when degenerate semiconductors are used as the partner layers.^[4] The value of V_{bi} of 1.02 eV has already been estimated for an n-FeSi₂/p-Si heterojunction solar cell.^[12] It has also been reported that V_{bi} increases for increasing doping concentrations.^[36] Although V_{bi} is higher than E_G in the current work, V_{OC} (958 mV) is within the V_{bi} maximum limit (1060 mV) for cell 5.

The BSF layer suppresses the carriers and reflects the electrons back before they reach the back contact of the solar cell. By preventing the electrons from diffusing and recombining at the rear metallic surface, this layer improves the charge collection chances. It also reduces the surface recombination

velocity (SRV) for the carriers at the rear side of the solar cell. By introducing BSF, we can minimize the SRV at the rear contact.^[21] Thus, the V_{OC} is improved accordingly. Furthermore, the V_{OC} in a BSF-aided cell such as cell 3 is at least 300 mV higher than that in the BSF-free solar cell of type 1, as shown in Figure 3b.

3.2.3. Effect on FF and PCE

The FFs in Figure 4a are due to the combined effect of the V_{OC} , J_{SC} , and those at maximum power point, V_{MPP} , J_{MPP} , corresponding to each cell type. Scientifically, FF can be written as Equation (4)^[20]

$$FF = \frac{V_{MPP} \times J_{MPP}}{V_{OC} \times J_{SC}} \quad (4)$$

The PCE of Figure 4b can be mathematically expressed as Equation (5)^[20]

$$PCE = \frac{V_{OC} \times I_{SC} \times FF}{P_{in}} \quad (5)$$

where P_{in} is the incident radiation power from sunlight. The enhanced V_{OC} therefore improves the PCE of the solar cell.

3.2.4. Effect on the Quantum Efficiency

The QE as a function of wavelength (λ) or photon energy ($h\nu$) is defined as the ratio of the number of carriers collected by the solar cell to the number of photons incident on that cell. QE equals 1 for a particular wavelength if all the incident photons are absorbed, and all the resultant carriers are collected. QE is 0 if photons have energy $h\nu$ below the E_G of the absorber. Again, the lower and higher λ spectra of the QE curve depend on the window and absorber layer diffusion length, respectively. To exhibit the effect of the thickness and BSF layer on the QE, solar cells for the cases of cells 1 and 3 were simulated.

The absorber thickness was varied from 50 to 300 nm in the simulation. In both cases, for the window (Si) and absorber (FeSi₂) layers, doping densities were kept the same at 10^{17} cm^{-3} , whereas it was 10^{19} cm^{-3} for the BSF layer. The thickness was taken as 100 nm for the window and 50 nm for the BSF. Figure 5 represents the simulation outcomes, where 'C-' denotes the cell number and numerical values in parenthesis indicate thicknesses in nanometers. Light with different frequencies (or wavelengths) is absorbed at different thicknesses of the absorber layer. For instance, the light-incident top surface absorbs lower wavelength photons, whereas the bulk material absorbs higher wavelength photons.^[35] FeSi₂ can absorb light in a wide range, from the visible to the NIR region up to 1675 nm.^[6] It is clear from Figure 5 that the absorber layer can be optimized easily with a thickness of 300 nm as it can absorb most of the incident photons. Figure 5 also shows that with a BSF layer, cell 3 captures photons more easily compared to cell 1 without a BSF. The BSF layer increases the spectral response of the cell in the longer wavelength portion,^[3] as shown in the same figure.

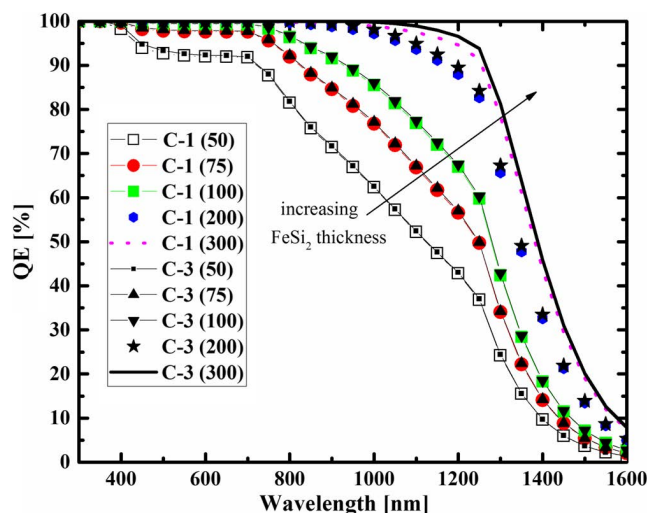


Figure 5. QE (λ) versus FeSi₂ thickness for solar cells 1 and 3.

The interface p⁺/p acts as a p/n junction, forming an electric field at that interface. This field initiates a barricade to the negatively charged carriers (minorities) flowing to back surface. Finally, the BSF causes a built-in-potential that essentially improves the charge collection probability, and V_{OC} . In other words, the blocking effect by the BSF layer improves the chances of minority carrier collection. The increased collection probability promotes the QE notably.

Taking the material cost and the cell performance into consideration, and looking at Figure 3–5, a thin layer of 300 nm is optimized for the FeSi₂ absorber layer for our proposed solar cells.

3.3. The Impact of the Window and BSF Layers' Thickness on Device Performance

In cells 1 and 2, the absorber partner layers do not contribute to efficiency increment. In cell 1, increasing the n-type Si layer thickness from 50 to 300 nm causes almost no effect on the output parameters, as shown in Figure 6a. Increasing the window

layer thickness up to 500 nm (Figure 6b) in cell 2 hinders the photons from reaching the absorber. However, this is minimal as the current decreases by 1 mA only. A thicker window will allow a smaller number of photons to be transmitted to the bulk absorber. Thus, the current will be reduced. Mainly, the p- and n-type Si layers in the pin structure of cell type 2 cause a compatible built-in potential to support the charge carrier separation.^[4] The 50 nm thin layer for both the Si window and the Si BSF layers was used for further study, until specified elsewhere in this article.

Figure 7 shows the effect of BaSi₂ BSF layer thickness on the PV parameters in cell 4. One can hardly find any improvements in the PV parameters by a thicker BSF, as shown in the figure. Therefore, it can be concluded that the BaSi₂ layer acts only as the BSF layer, not like an absorber layer.

The current density, J_{SC} , is almost constant at 51.67 mA cm⁻² as this layer does not contribute to significant photon absorption. Similarly, all other outputs are approximately the same as 715 mV, 85.46%, and 31.61% for V_{OC} , FF, and η , respectively. The analogous influence of the BSF layer was reported in the literature. However, the thicker the BSF layer, the more photons with longer wavelengths are absorbed, generating more EHPs.^[3] We can identify the positive effect of using a BSF layer in FeSi₂-based solar cells by comparing them with cells 1 and 2 having no BSF layer.

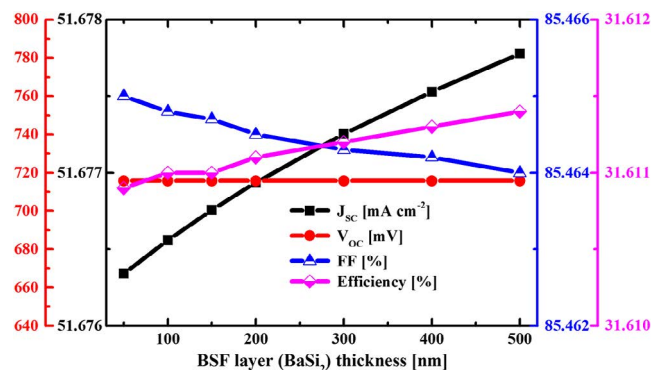


Figure 7. The output parameters versus BaSi₂ thickness in cell 4.

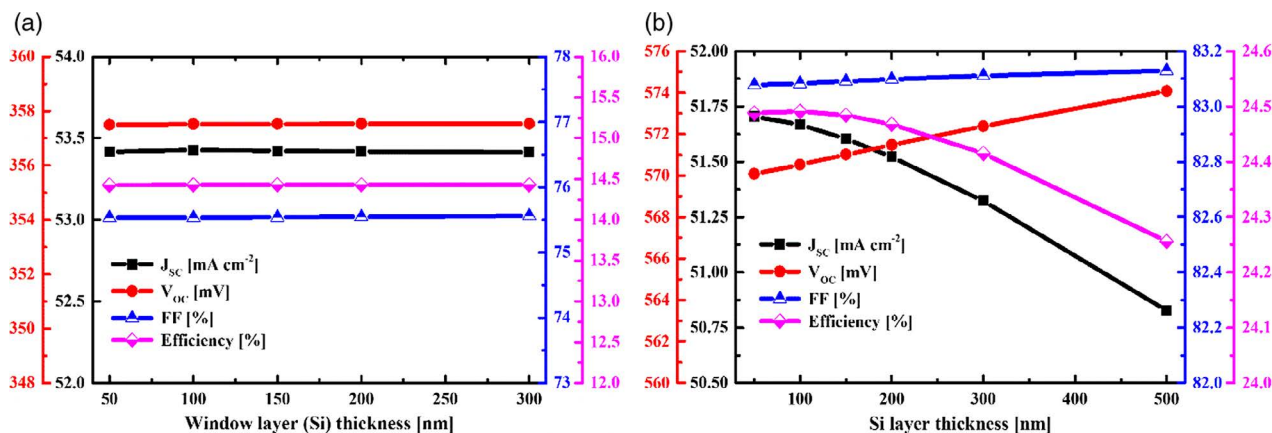


Figure 6. The PV parameters versus Si thickness for a) cell 1 and b) cell 2.

As a window layer, the effect of CdS is more pronounced in cell 5 compared to the other windows used in this study, which is confirmed from the discussion in Section 3.2. CdS material has a bandgap of 2.4 eV, which implies that photon energies > 2.4 eV can be absorbed by this layer in the lower wavelength (or higher frequency) range from 270 to 520 nm.^[30] For this reason, if the CdS window layer is thick enough, photons in a lower wavelength range will not be passed to the absorber. As a result, the output current will be lower compared to a thinner window. A similar result was obtained for the current and voltage.^[37] Figure 8 shows how a thicker CdS layer reduces the J_{SC} and V_{OC} .

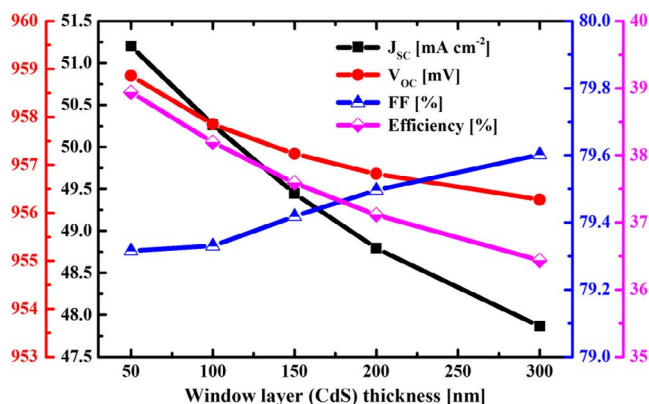


Figure 8. The output parameters versus CdS window layer thickness in cell 5.

Considering all consequences, the thicknesses of the BaSi₂ BSF and CdS window layers were chosen to 50 nm in the device simulation.

3.4. Effect of Doping Density

For analyzing the doping effect in the modeled solar cells, cell 1 was studied first. The acceptor impurity density, N_A in the 1 μm thin FeSi₂ absorber was varied from 10^{14} to 10^{19} cm⁻³. The donor type doping was fixed at 10^{17} cm⁻³ in the 50 nm thin Si window at 300 K. The effect is shown in Figure 9a.

With increasing N_A , J_{SC} reduced drastically. In contrast, V_{OC} increased from 350 to 452 mV. The key reason behind this is that the saturation current and thus V_{OC} increase with increasing N_A . But at higher N_A , the recombination rate is higher, which minimizes the collection probability of the photogenerated electrons, and the J_{SC} .^[20] Efficiency and FF also increase with increasing N_A , as shown in Figure 9a. A similar result for these parameters was obtained.^[20]

Figure 9b shows the donor ion doping effect in 50 nm thin Si window layer of cell 1 at 300 K. The acceptor density in the FeSi₂ absorber was kept constant at 10^{18} cm⁻³. Adding donor ions improves the electrical conductivity in the window layer by the addition a lot of free electrons.^[37] Thus, performance improves in the cell due to the increase in the electron current.

From the previous discussion and considering the performance, cost, and growth rate, 10^{18} cm⁻³ was opted for both layers in cell 1.

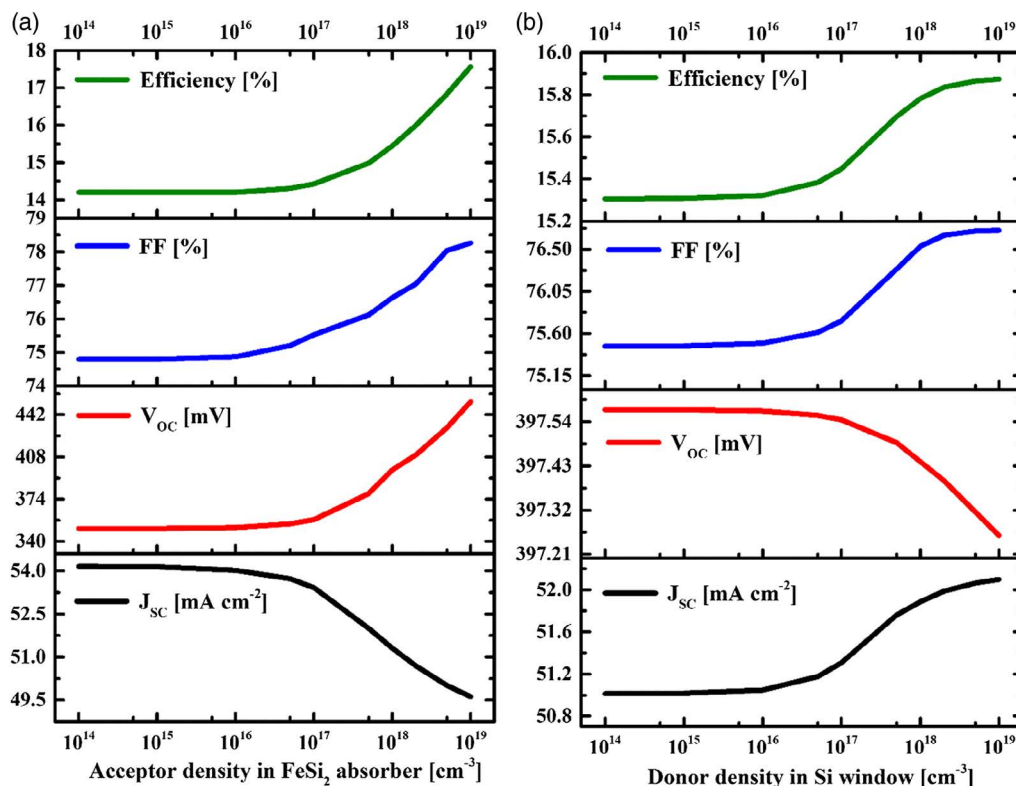


Figure 9. The effect of doping density on the PV parameters in cell 1: a) N_A in FeSi₂ and b) N_D in Si.

For solar cell 2 at 300 K, we varied the doping density (10^{14} to 10^{20} cm^{-3}) simultaneously in 50 nm thin p-Si and 50 nm thin n-Si layers, taking intrinsic doping ($N_A = N_D = 0 \text{ cm}^{-3}$) in the 300 nm thin absorber layer of the pin cell. V_{OC} increases by $\approx 80 \text{ mV}$, efficiency increases by $\approx 3\%$, and FF is almost stable at 83%, whereas J_{SC} decreases by $\approx 1 \text{ mA cm}^{-2}$ when both densities are simultaneously varied in the said range. For solar cell 3 at 300 K, taking 10^{18} cm^{-3} as the donor density in the 50 nm thin n-Si window layer and $5 \times 10^{19} \text{ cm}^{-3}$ as the acceptor density in the 50 nm thin p^+ -Si BSF layer, we varied N_A from 10^{14} to 10^{20} cm^{-3} in the 300 nm thin p-FeSi₂ absorber layer. From N_A (absorber) of 10^{14} to 10^{19} cm^{-3} , all the parameters give approximately similar results, with a little bit of enhancement after 10^{17} cm^{-3} . In cell 4, the doping was 10^{19} cm^{-3} for the p^+ -BaSi₂ layer. The same doping variation in the absorber layer was performed for solar cells 4 and 5, which gave results identical to those of cell 3.

Having a peak defect density of $10^{14} \text{ eV}^{-1} \text{ cm}^{-3}$ in all layers, the optimized doping density (cm^{-3}) is 10^{18} (p-n) for cell 1; 10^{17} (p-n) for cell 2; 10^{18} (p-n) and 5×10^{19} (p^+) for cell 3; 10^{18} (n), 10^{17} (p), and 10^{19} (p^+) for cell 4; and 10^{18} (n), 10^{17} (p), and 10^{19} (p^+) for cell 5.

3.5. The Impact of Bulk Defect Density on Device Structures

The recombination of charge carriers must take place somewhere within the cell. In the bulk semiconductor material, we can introduce recombination through defects such as neutral, donor, acceptor, and amphoteric. defects that mostly contribute to the Shockley–Read–Hall (SRH), Auger, or radiative recombination. In this work, we used single-acceptor and single-donor-type bulk defects in n- and p-type materials for all solar cells, whereas both defects were introduced in the intrinsic FeSi₂ layer of cell 2.

The absorber layer peak defect density was varied from 10^{12} to $10^{19} \text{ eV}^{-1} \text{ cm}^{-3}$ taking a constant peak defect density of $10^{14} \text{ eV}^{-1} \text{ cm}^{-3}$ for all other layers. The dominant effect of bulk defects present in the FeSi₂ absorber layer on the PV parameters is shown in **Figure 10**. A peak defect density of about $10^{14} \text{ eV}^{-1} \text{ cm}^{-3}$ or less in the absorber layer is tolerable for achieving an efficiency of $\approx 28\%$, $\approx 31\%$, and $\approx 38\%$ in solar cells 3–5, respectively.

All the PV parameters excluding the J_{SC} degrade significant with increasing peak defect densities starting from $10^{15} \text{ eV}^{-1} \text{ cm}^{-3}$. This degradation is mainly due to the higher

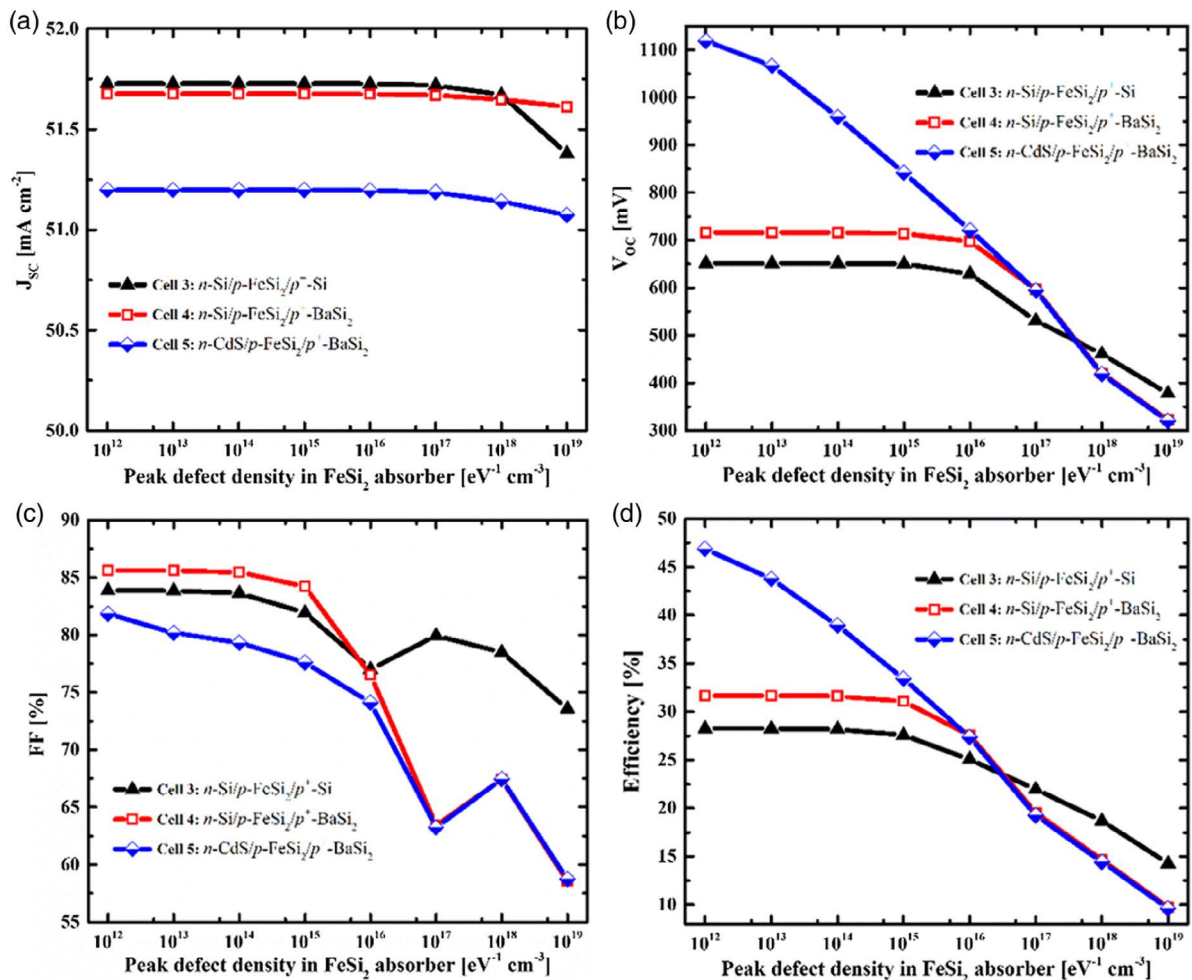


Figure 10. The effect of peak defect density on the PV parameters: a) J_{SC} , b) V_{OC} , c) FF, and d) PCE of the proposed solar cells.

recombination rate in the absorber when the defect density is higher. At a high defect density, the SRH recombination is the dominant recombination factor, which is confirmed by the G–R profile of the simulator. As an outcome, less photogenerated charge carriers are obtained, reducing the output accordingly. The SRH recombination generally reduces the V_{OC} to the highest degree, thus reducing the FF and the efficiency. Recombination loss of photon-energized charge carriers immediately affects the solar cell PCE.^[21] The device is greatly hampered by the presence of interface and bulk defects, which reduce the minority carrier lifetime and diffusion length.^[6] At the maximum incorporated defects of $10^{19} \text{ eV}^{-1} \text{ cm}^{-3}$, solar cells 3–5 show the least efficiencies, $\approx 14.3\%$, $\approx 9.7\%$, and $\approx 9.6\%$, respectively. Presently, the highest practical lab efficiency of an FeSi_2 -based solar cell is 3.7% .^[7] It can be stated that the experimentally grown films lack purity or good quality. Lessening of the number of defects was said to be the most essential task in attaining a high PCE.^[4] Thus, to have the required and potential outcomes from FeSi_2 -based solar cells, we need to grow high-quality FeSi_2 films and apply the interfacial engineering design.

3.6. Shift in PV Parameters with the Variation of Temperature

The deviation of PV parameters of the optimized heterojunction solar cells with respect to the operating temperature (T) is shown in **Figure 11**. The change in output was measured for temperatures over 100 K in the range 273–373 K. The J_{SC} is almost independent (slight increase) of T , as shown in Figure 11. The operating temperature mainly affects the E_g of the materials. As E_g decreases and more photons have adequate energy to produce EHPs, J_{SC} increases slightly with T , which is still a minor effect on J_{SC} .

Similarly to the absorber layer defects, high temperature reduces the photogenerated voltage noticeably. The change in V_{OC} was lower in cell type 5 than in either cell 3 or cell 4. Increasing the operating temperature increases the electrons' thermal energy in the cells. The flow of free charge carriers is obstructed by the lattice vibrations, causing the junction to lose its power. Thus, the efficiency reduces because of the collision with thermally energized atoms.^[37] With higher energies, electrons recombine with holes timelier. The recombination rate is raised likewise. This reduces the saturation current while maximizing the reverse saturation current.^[24]

In addition, the bond model of a semiconductor bandgap reveals that the increased temperature reduces the energy gap by reducing the bond energy. As a consequence, V_{OC} decreases with the increase in T . This eventually diminishes the FF and PCE of the solar cells.

As shown in Figure 11, it can be concluded that higher temperatures strongly dominate the devices compared to the lower ones. The temperature coefficient, T_C , of the conversion efficiency in $\% \text{ K}^{-1}$ can be defined at standard test conditions (STC) as Equation (6)^[15,38]

$$T_C = \left(\frac{1}{\eta_{STC}} \frac{d\eta}{dT} \times 100 \right) [\% \text{ K}^{-1}] \quad (6)$$

where η_{STC} is the cell efficiency at STC at 298 K and η is the efficiency at temperature T greater than 298 K. Using Equation (6),

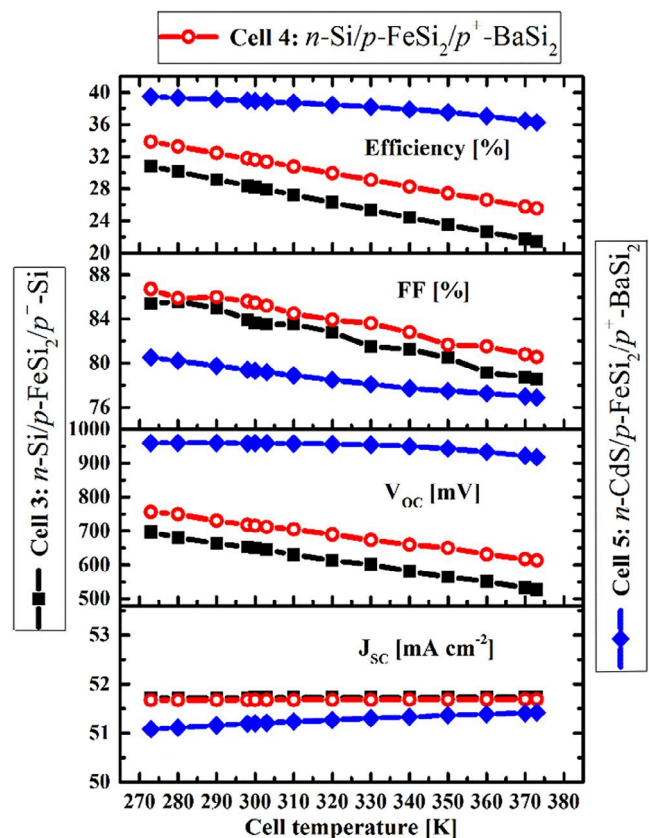


Figure 11. Variation of cell output parameters with respect to the temperature.

the calculated temperature coefficients for solar cells 3 and 4 are found to be $-0.3261\% \text{ K}^{-1}$ and $-0.2619\% \text{ K}^{-1}$, respectively. The structure of cell 5 offers the least possible T_C of $-0.0927\% \text{ K}^{-1}$, proclaiming the highest thermal stability among the three cases. In addition, the temperature coefficients of the BSF-connected

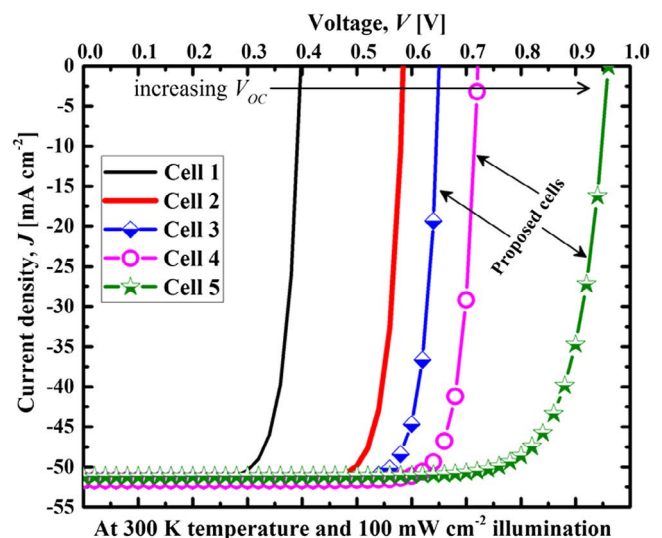


Figure 12. The illuminated J – V characteristics of the optimized solar cells.

solar cells 3–5 are much lower than that of solar cell 1 ($-0.5813\% \text{ K}^{-1}$) and 2 ($-0.3412\% \text{ K}^{-1}$) modeled without the BSF layer.

3.7. Summary Results of the Studied Cases

Figure 12 shows the illuminated J – V characteristics of the five optimized heterojunction solar cells we modeled. **Table 2** summarizes the respective PV parameters of the corresponding cells.

A comparison of the designed solar cell structures with reported ones is provided in **Table 3**, which also shows how the development occurs for FeSi_2 -based solar cells. The PCEs obtained experimentally start from 0.22% to the highest efficiency of 3.7%, although the practical efficiency could be further improved up to 6% by the use of an antireflection coating (ARC) layer to reduce 40% reflectance of the visible light from FeSi_2 .^[1,14] The previous theoretical approaches show how the efficiency increases from 7% to 27.9%, till date. In short, we

achieved better V_{OC} and PCE than the reported values.^[4–7,10–14] Obtained FF is higher than that in most of the reported value, whereas it is competitive with other solar cells.^[4,5] The J_{SC} realized in this study is less only than in Shaban et al.,^[6] but higher than all other reported values.^[4,5,7,10–14] The current study provides a design guideline to achieve the utmost efficiency of $\approx 39\%$ from the three layered FeSi_2 -based npp⁺ heterojunction solar cells.

4. Conclusions

Five distinct npp⁺ heterojunction solar cells based on the eco-friendly FeSi_2 absorber have been modeled to harness the visible and NIR light and studied in detail. Simulations have been carried out using SCAPS-1D simulator to investigate the effects of layer thickness, doping and defect densities, band offsets as well as temperature on the PV performance of the devices. In regard to the PCE and V_{OC} , the proposed three-layered npp⁺ devices are superior to the two-layered np (or pn) and three-layered

Table 2. Summary of the optimized PV parameters in current study at 300 K.

No.	Studied solar cell type	FeSi_2 depth [nm]	J_{SC} [mA cm^{-2}]	J_{MPP} [mA cm^{-2}]	V_{OC} [mV]	V_{MPP} [mV]	FF [%]	PCE [%]
1	Cell 1: n-Si/p- FeSi_2	1000	51.887	47.834	397.440	329.932	76.530	15.782
2	Cell 2: p-Si/i- FeSi_2 /n-Si	300	51.710	49.188	585.491	507.520	82.455	24.964
3	Cell 3: n-Si/p- FeSi_2 /p ⁺ -Si	300	51.727	49.418	651.282	570.264	83.651	28.181
4	Cell 4: n-Si/p- FeSi_2 /p ⁺ -BaSi ₂	300	51.676	49.604	715.743	637.256	85.465	31.610
5	Cell 5: n-CdS/p- FeSi_2 /p ⁺ -BaSi ₂	300	51.197	47.931	958.860	812.354	79.316	38.937

Table 3. Performance comparison of our optimized results with the reported values (Exp. = Experimental, Theo. = Theoretical. In the 'Junction formed' column, 'nc' and 'a' denote nanocrystalline and amorphous phase, respectively).

No.	Research type	Junction formed	Device thickness/active area	FeSi_2 depth [nm]	J_{SC} [mA cm^{-2}]	V_{OC} [mV]	FF [%]	PCE [%]	Ref.
1	Exp.	n-nc- FeSi_2 /p-Si	1.19 cm ²	350	3.8	160	38	0.22	[11]
2	Exp.	n- FeSi_2 /p-Si	$\approx 261 \mu\text{m}$	500	12.81	176	28	0.63	[12]
3	Exp.	$\text{FeSi}_2 + \text{Al}/\text{Al}/\text{Si}$	5 × 5 mm ²	50	3.91	450	≈ 46	0.81	[13]
4	Exp.	p- FeSi_2 (Al)/n-Si	2.5 × 2.5 mm ²	≈ 80	4.2	450	44	0.82	[14]
5	Exp.	n- FeSi_2 /p-Si	4 × 4 mm ²	230	14.84	450	55	3.7	[7]
6	Theo.	n- FeSi_2 /p-Si	3.15 μm	150	≈ 46	391	73	7–18	[10]
7	Theo.	n- FeSi_2 /p-Si	$\approx 300 \mu\text{m}$	1–10	34–39	≈ 612	≈ 80	16–19	[5]
8	Cell 1	n- FeSi_2 /p-Si	1.05 μm	1000	51.88	≈ 397	76.53	15.78	a)
9	Theo.	p- FeSi_2 /n-Si	3.15 μm	150	–	–	–	14	[10]
10	Theo.	p- FeSi_2 /n-Si	–	300	≈ 44	500	80	17.7	[4]
11	Theo.	n- FeSi_2 /p-Si	–	300	≈ 43	≈ 400	77	13.3	[4]
12	Theo.	p-Si/i- FeSi_2 /n-Si	700 nm	300	≈ 43	≈ 690	84	24.7	[4]
13	Cell 2	p-Si/i- FeSi_2 /n-Si	400 nm	300	51.71	≈ 585	82.45	24.96	a)
14	Theo.	n-a-Si/i- FeSi_2 /p-a-Si	400 nm	300	≈ 40	≈ 900	77.7	27.9	[4]
15	Theo.	p-nc-Si/i-nc- FeSi_2 /n-nc-Si	500 nm	300	53	590	79	25.2	[6]
16	Cell 3	n-Si/p- FeSi_2 /p ⁺ -Si	400 nm	300	51.72	≈ 651	83.65	28.18	a),b)
17	Cell 4	n-Si/p- FeSi_2 /p ⁺ -BaSi ₂	400 nm	300	51.67	≈ 716	85.46	31.61	a),b)
18	Cell 5	n-CdS/p- FeSi_2 /p ⁺ -BaSi ₂	400 nm	300	51.19	≈ 958	79.31	38.94	a),b)

a) Indicates current work; b) Indicates proposed cell.

pin (or nip) devices. Simulation results reveal that the best PV performances are exhibited by cell 5 (n-CdS/p-FeSi₂/p⁺-BaSi₂), whereas cell 4 (n-Si/p-FeSi₂/p⁺-BaSi₂) and cell 3 (n-Si/p-FeSi₂/p⁺-Si) hold the second and third position, respectively. Additionally, cell 5 offers the highest thermal stability as revealed by the obtained temperature coefficients. When the absorber's peak defect density exceeds 10¹⁵ eV⁻¹ cm⁻³, all the parameters excluding the current deteriorate drastically. A thin layer of 300 nm is optimized for the FeSi₂ absorber with 50 nm thin adjacent layer(s) in all cases, except cell 1 has a 1 μm thick absorber. With a peak defect density of 10¹⁴ eV⁻¹ cm⁻³ in all layers, the highest V_{OC} of 958 mV and PCE of ≈39% are theoretically realized for the FeSi₂-based solar cells. In comparison to the recent technologies, with a device thickness of 400 nm, the current work leads toward an ultrathin high-efficiency 3G solar cell with cost effectiveness for large-scale applications in space and multikilowatt systems of PV industries.

Acknowledgements

The authors highly appreciate Professor Marc Burgelman of the Department of Electronics and Information Systems of the University of Gent, Belgium, for providing the SCAPS software package (Version: 3.3.07).

Conflict of Interest

The authors declare no conflict of interest.

Keywords

back-surface field, iron disilicide, numerical simulations, photovoltaics, third-generation solar cells

Received: November 8, 2019

Revised: January 19, 2020

Published online: February 13, 2020

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